TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
Question Set: A

| | | | |
|---|---|---|---|
| | | Marks | |
| **Level** | BE | **Pass Marks** | 24 |
| **Program** | BCT | **Time** | 3 hrs. |
| **Year/Part** | III/I | | |

**Subject: Advanced Python Programming for Data Science (ENCT 325)**

- *Candidates are required to give their answers in their own words as far as practicable.*
- *Attempt All Questions*
- *Assume suitable data if necessary*
- *The figures in the margin indicate Full Marks*

| QN | Questions | Marks | Units |
|---|---|---|---|
| 1 | **a)** Explain the principles of Object-Oriented Programming (OOP) in Python. Write a Python class DataProcessor with attributes for dataset name and data, and a method to display a summary. Extend it with a subclass StudentProcessor that overrides the summary method to include subject-wise mean score.<br><br>**b)** What is a Python decorator? Write a Python decorator log_execution that prints the function name and execution time before and after a function call. Demonstrate it on a sample data-loading function. | 3+3 | 1 |
| 2 | **a)** What is exception handling in Python? Explain the use of try, except, else, and finally blocks with a coding example that handles FileNotFoundError and ValueError during data file loading.<br><br>**b)** Write Python code to read a CSV file containing student records and export the data as a JSON file. Handle the case where the CSV file may not exist or contain malformed rows. | 3+3 | 1+2 |
| 3 | **a)** Explain the concept of a REST API and how Python's requests library is used to interact with one. Write Python code to retrieve student data from a sample API endpoint (https://api.example.com/students), parse the JSON response, and display only the StudentID and Score fields.<br><br>**b)** Explain web scraping using BeautifulSoup. Write Python code to scrape a table of student rankings from a sample HTML page, extract all rows, and store the data in a Pandas DataFrame. | 3+3 | 2 |
| 4 | **a)** Explain the difference between wide-format and long-format data. When would you convert from one format to the other in a data science workflow?<br>**b)** Consider the following student performance dataset: | 2+4 | 3 |

| StudentID | Subject | Score | Grade | AttendancePct | StudyHours |
|---|---|---|---|---|---|
| S01 | Math | 82 | B | 90 | 5 |
| S02 | Science | NaN | A | 85 | 6 |
| S03 | Math | 75 | C | NaN | 4 |
| S04 | English | NaN | B | 78 | NaN |
| S05 | Science | 91 | A | 92 | 7 |
| S06 | English | NaN | C | 70 | 3 |

| | | | |
|---|---|---|---|
| | Write Python code to: load the dataset, handle all missing values using an appropriate strategy for each column type, and create a pivot table showing the average Score per Subject. | | |
| 5 | **a)** Describe at least three approaches for memory optimization in Python when processing large datasets. How do chunking and appropriate data types reduce memory usage?<br><br>**b)** Explain the stages of a data-cleaning pipeline. Using the dataset from Question 4(b), write Python code to build a reusable pipeline that: fills missing numeric values with the column median, applies Min-Max scaling to Score and StudyHours, and applies label encoding to Subject and Grade. | 2+4 | 3 |
| 6 | **a)** Explain skewness and kurtosis. How do these measures help in understanding the shape and tail behaviour of a data distribution?<br><br>**b)** Using the dataset from Question 4(b), write Python code to compute the skewness and kurtosis of the Score column after handling missing values, and interpret the results. | 2+3 | 4 |
| 7 | A school board claims that the average student score is 75. A random sample of 30 students produces the following statistics:<br>Sample mean = 71.4<br>Sample standard deviation = 9.2<br>Significance level = 0.05<br><br>Formulate the null and alternative hypotheses, compute the test statistic manually, state the decision rule (t-critical for df = 29 at $\alpha = 0.05$ is $\pm 2.045$), and write Python code using scipy.stats to perform the one-sample t-test and print the conclusion. | 5 | 4 |
| 8 | **a)** Explain the principles of effective data visualization. What makes a visualization clear, accurate, and actionable?<br><br>**b)** Using the dataset from Question 4(b), write Python code using Matplotlib to create a figure with two subplots side by side: a histogram showing the distribution of Score on the left, and a box plot showing Score variation across Subjects on the right. Handle missing values before plotting and apply appropriate labels, titles, and a figure title. | 2+4 | 5 |
| 9 | **a)** Explain interactive visualization using Plotly. Using the dataset from Question 4(b), write Python code to create a Plotly 3D scatter plot showing the relationship among StudyHours, AttendancePct, and Score, with each point coloured by Subject.<br><br>**b)** Explain the concept of an ETL (Extract, Transform, Load) pipeline and its role in data engineering. Write Python code to illustrate a simple ETL workflow that extracts data from a CSV file, transforms it by renaming columns and filtering rows with Score > 70, and loads the result into a new CSV file. | 3+3 | 5+6 |
| 10 | **a)** Write and explain a CRON expression to run a Python analytics script every day at 8:00 AM. Describe each field of the expression and explain how the schedule library in Python can be used as an alternative to CRON for simple task scheduling.<br><br>**b)** What is automated report generation in data engineering? Why is it important in production data pipelines?<br><br>**c)** Using the dataset from Question 4(b), write Python code to generate an Excel report containing a summary sheet with descriptive statistics (mean, median, std, min, max) of Score, AttendancePct, and StudyHours using openpyxl or xlsxwriter. | 3+2+3 | 6 |

**TRIBHUVAN UNIVERSITY**
**INSTITUTE OF ENGINEERING**
**Examination Control Division**
Question Set: B

| Exam | Regular | Full Marks | 60 |
|---|---|---|---|
| Level | BE | Pass Marks | 24 |
| Program | BCT | Time | 3 hrs. |
| Year/Part | III/I | | |

**Subject: Advanced Python Programming for Data Science (ENCT 325)**

- *Candidates are required to give their answers in their own words as far as practicable.*
- *Attempt All Questions*
- *Assume suitable data if necessary*
- *The figures in the margin indicate Full Marks*

| QN | Questions | Marks | Units |
|---|---|---|---|
| 1 | **a)** Explain Python generators and how they differ from regular functions and iterators. Write a Python generator function data_stream that yields records one at a time from a large list of order dictionaries, simulating memory-efficient data streaming in a data science pipeline.<br><br>**b)** Explain the use of lambda expressions and the map() and filter() functions in Python. Write code to use these on a list of order prices: apply a 10% discount using map(), then filter to retain only orders with a discounted price above 1000. | 3+3 | 1 |
| 2 | **a)** Explain PEP-8 coding standards and their importance in collaborative Python projects. Rewrite the following poorly formatted code snippet following PEP-8 conventions:<br><br>`def calcTOTAL(p,q,d):return(p*q)*(1-d)`<br>`x=calcTOTAL(500,3,0.1);print('total=',x)`<br><br>**b)** Explain how Python's SQLAlchemy library is used to connect to a relational database and perform queries. Write Python code to connect to a SQLite database, create an Orders table, insert three records, and retrieve all orders where Price is greater than 5000. | 3+3 | 1+2 |
| 3 | **a)** Explain how Python handles database access for both relational (SQLite) and non-relational (MongoDB) databases. Write Python code using sqlite3 to query a sales orders table and return all records where Category is 'Electronics', sorted by Price descending.<br>**b)** What is chunking in data processing? Write Python code using Pandas read_csv() with chunksize to process a large orders CSV file in chunks of 1000 rows, accumulating the total Revenue per Category across all chunks. | 3+3 | 2 |
| 4 | **a)** Explain the difference between merge and join operations in Pandas. When would you use an inner join versus an outer join in a data science workflow?<br>**b)** Consider the following e-commerce orders dataset: | 2+4 | 3 |

| OrderID | Category | Price | Quantity | Rating | DeliveryDays |
|---|---|---|---|---|---|
| O01 | Electronics | 15000 | 2 | 4.5 | 3 |
| O02 | Clothing | NaN | 5 | 3.8 | 5 |
| O03 | Electronics | 22000 | 1 | NaN | 2 |
| O04 | Clothing | 1800 | NaN | 4.1 | NaN |
| O05 | Books | 450 | 8 | 4.7 | 4 |
| O06 | Books | NaN | 3 | 3.9 | 6 |

| | | | |
|---|---|---|---|
| | Write Python code to: load the dataset, handle all missing values using an appropriate strategy for each column, and create a pivot table showing the average Price per Category. | | |
| 5 | a) What is a reusable data-cleaning pipeline? Explain its components and how it improves reproducibility in data science projects.<br><br>b) Using the dataset from Question 4(b), write Python code to build a reusable cleaning pipeline that: imputes missing numeric values using the column median, applies StandardScaler to Price, Quantity, and Rating, and applies one-hot encoding to Category. The pipeline should be implemented as a function that can be reused on any new batch of data. | 2+4 | 3 |
| 6 | a) Explain correlation and covariance. How are they used in EDA to understand relationships between variables? What is the difference between Pearson and Spearman correlation?<br><br>b) Using the dataset from Question 4(b), write Python code to compute the correlation matrix for Price, Quantity, Rating, and DeliveryDays after handling missing values, and display it as a Seaborn heatmap with annotations. | 2+3 | 4 |
| 7 | An e-commerce platform claims that the average delivery time is 5 days. A random sample of 40 orders yields the following statistics:<br>Sample mean = 5.8 days<br>Sample standard deviation = 2.1 days<br>Significance level = 0.05<br><br>Formulate the null and alternative hypotheses, compute the t-statistic manually, state the decision rule (t-critical for df = 39 at $\alpha = 0.05$ is $\pm 2.023$), and write Python code using scipy.stats.ttest_1samp to perform the test and print the p-value and conclusion. | 5 | 4 |
| 8 | a) What are the key principles of dashboard design for data storytelling? How should chart types be selected to communicate comparison, distribution, relationship, and composition data?<br><br>b) Using the dataset from Question 4(b), write Python code using Matplotlib to create a figure with two subplots: a bar chart showing the average Price per Category on the left, and a scatter plot of Price versus Rating on the right. Handle missing values before plotting and add appropriate labels and a figure title. | 2+4 | 5 |
| 9 | a) Explain Seaborn's use in statistical visualization. Using the dataset from Question 4(b), write Python code to create a Seaborn heatmap of the correlation matrix for all numeric columns and a pair plot of Price, Quantity, Rating, and DeliveryDays, with points coloured by Category.<br>b) Explain error handling and logging in ETL pipelines. Write Python code implementing an ETL pipeline that logs each stage (extract, transform, load) to a log file using Python's logging module, and raises and catches a custom exception DataQualityError if more than 20% of rows contain missing values. | 3+3 | 5+6 |
| 10 | a) Write a CRON expression to run a Python pipeline every Monday at 7:00 AM and explain each field. How does the schedule library differ from CRON for production scheduling?<br><br>b) Explain logging and monitoring in data pipelines. What information should be captured at each pipeline stage and why?<br><br>c) Using the dataset from Question 4(b), write Python code to generate an automated HTML report that includes a summary statistics table for Price, Rating, and DeliveryDays and a count of orders per Category. Save the report as report.html. | 3+2+3 | 6 |

**TRIBHUVAN UNIVERSITY**
**INSTITUTE OF ENGINEERING**
**Examination Control Division**
Question Set: C

| | | | |
|---|---|---|---|
| **Exam** | Regular | **Full Marks** | 60 |
| **Level** | BE | **Pass Marks** | 24 |
| **Program** | BCT | **Time** | 3 hrs. |
| **Year/Part** | III/I | | |

**Subject: Advanced Python Programming for Data Science (ENCT 325)**

- *Candidates are required to give their answers in their own words as far as practicable.*
- *Attempt All Questions*
- *Assume suitable data if necessary*
- *The figures in the margin indicate Full Marks*

<table>
<tr><th>QN</th><th>Questions</th><th>Marks</th><th>Units</th></tr>
<tr><td>1</td><td>a) Explain Python iterators and the iterator protocol (__iter__ and __next__). Write a Python class RecordIterator that iterates over a list of patient records and yields one record at a time. Demonstrate its use in a data science context.<br><br>b) Explain the collections module in Python. Demonstrate the use of defaultdict and Counter with examples relevant to a hospital data scenario (e.g., counting patients per ward, grouping records by diagnosis).</td><td>3+3</td><td>1</td></tr>
<tr><td>2</td><td>a) Explain encapsulation and inheritance in OOP with Python. Write a Patient class with private attributes (patientID, age, ward) and getter/setter methods. Extend it with a CriticalPatient subclass that adds a priority_level attribute and overrides a display_info() method.<br><br>b) Explain how Python reads data from a JSON file and how it can connect to a non-relational database such as MongoDB using pymongo. Write Python code to load a JSON file of patient records and insert all records into a MongoDB collection named patients.</td><td>3+3</td><td>1+2</td></tr>
<tr><td>3</td><td>a) Compare REST and SOAP APIs in terms of architecture, data format, and use cases. Write Python code using the requests library to send a GET request to a hospital API endpoint, handle potential HTTP errors, and extract all records where Ward is 'ICU'.<br><br>b) Explain web scraping with BeautifulSoup. Write Python code to scrape a table of hospital ward statistics from an HTML page, parse all rows into a list of dictionaries, and convert the result to a Pandas DataFrame.</td><td>3+3</td><td>2</td></tr>
<tr><td>4</td><td>a) Explain reshaping and pivoting in Pandas. What is the difference between pivot() and pivot_table()? When is melt() used?<br>b) Consider the following hospital patient dataset:
<table>
<tr><th>PatientID</th><th>Ward</th><th>Age</th><th>BloodPressure</th><th>Cholesterol</th><th>LengthOfStay</th></tr>
<tr><td>P01</td><td>ICU</td><td>55</td><td>130</td><td>NaN</td><td>7</td></tr>
<tr><td>P02</td><td>General</td><td>42</td><td>NaN</td><td>210</td><td>4</td></tr>
<tr><td>P03</td><td>ICU</td><td>68</td><td>145</td><td>240</td><td>NaN</td></tr>
<tr><td>P04</td><td>General</td><td>NaN</td><td>120</td><td>195</td><td>3</td></tr>
<tr><td>P05</td><td>Surgical</td><td>60</td><td>NaN</td><td>230</td><td>8</td></tr>
<tr><td>P06</td><td>Surgical</td><td>NaN</td><td>135</td><td>NaN</td><td>6</td></tr>
</table></td><td>2+4</td><td>3</td></tr>
</table>

| | | | |
|---|---|---|---|
| | Write Python code to: load the dataset, handle all missing values using an appropriate strategy for each column type, and create a pivot table showing the average Cholesterol per Ward. | | |
| 5 | a) Explain how time-series data differs from cross-sectional data in terms of handling and preprocessing. What are the common challenges when working with datetime columns in Pandas?<br><br>b) Using the dataset from Question 4(b), write Python code to build a feature transformation pipeline that: encodes Ward using one-hot encoding, applies Z-score (StandardScaler) normalization to Age, BloodPressure, and Cholesterol, and outputs a clean transformed DataFrame ready for modelling. | 2+4 | 3 |
| 6 | a) Explain probability sampling methods used in data science: simple random, stratified, and systematic sampling. When is stratified sampling preferred over simple random sampling?<br><br>b) Using the dataset from Question 4(b), write Python code to perform stratified sampling — selecting 50% of patients from each Ward — after handling missing values. Print the resulting sample and verify the proportion per Ward. | 2+3 | 4 |
| 7 | A hospital administrator claims that the average length of stay is 5 days. A random sample of 28 patients produces the following statistics:<br>Sample mean = 6.3 days<br>Sample standard deviation = 2.4 days<br>Significance level = 0.05<br><br>Formulate the null and alternative hypotheses, compute the t-statistic manually, state the decision rule (t-critical for df = 27 at $\alpha = 0.05$ is $\pm 2.052$), and write Python code using scipy.stats.ttest_1samp to perform the test and print the p-value and your conclusion. | 5 | 4 |
| 8 | a) Explain data visualization storytelling. How does the choice of chart type support the narrative of a data-driven insight? Give two examples of appropriate chart types for medical data analysis.<br><br>b) Using the dataset from Question 4(b), write Python code using Matplotlib to create a figure with two subplots: a histogram showing the distribution of Age on the left, and a box plot showing BloodPressure variation across Wards on the right. Add clear titles, labels, and a main figure title. | 2+4 | 5 |
| 9 | a) Explain interactive visualization using Plotly. Using the dataset from Question 4(b), write Python code to create a Plotly interactive scatter plot of Age versus Cholesterol, with points coloured by Ward and sized by LengthOfStay.<br><br>b) Explain data storage and retrieval strategies used in data engineering pipelines. Compare storing processed data as CSV, Parquet, and in a relational database in terms of speed, compression, and query support. | 3+3 | 5+6 |
| 10 | a) Explain the stages of an ETL pipeline in data engineering. Write Python code to implement a complete ETL pipeline for the hospital dataset: extract from a CSV file, transform by handling missing values and encoding Ward, and load the result into a SQLite database table named clean_patients.<br><br>b) Write a CRON expression to run a weekly data refresh every Sunday at midnight (00:00). Explain each field of the expression.<br><br>c) Using the dataset from Question 4(b), write Python code to generate an automated PDF report containing the title, the date of generation, and a summary statistics table for Age, Cholesterol, and LengthOfStay using the reportlab or fpdf library. | 3+2+3 | 6 |

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| **TRIBHUVAN UNIVERSITY**<br>**INSTITUTE OF ENGINEERING**<br>**Examination Control Division**<br>Question Set: D | |

| Exam | Regular | Full Marks | 60 |
|---|---|---|---|
| Level | BE | Pass Marks | 24 |
| Program | BCT | Time | 3 hrs. |
| Year/Part | III/I | | |

**Subject: Advanced Python Programming for Data Science (ENCT 325)**

- *Attempt All Questions*
- *Assume suitable data if necessary*

| QN | Questions | Marks | Units |
|---|---|---|---|
| 1 | a) Explain Python decorators with a practical data science example. Write a decorator validate_dataframe that checks whether the input to a function is a non-empty Pandas DataFrame and raises a TypeError with a descriptive message if it is not.<br><br>b) Explain exception handling and logging in Python. Write a Python function load_weather_data(filepath) that: uses try-except to handle FileNotFoundError and pd.errors.ParserError, logs each event (info on success, error on failure) to a file weather.log using the logging module. | 3+3 | 1 |
| 2 | a) Explain how Python modules and packages are structured and created. Write a Python module weather_utils.py containing two functions: compute_stats(df, col) that returns mean, median and std of a column, and flag_missing(df) that prints the count of missing values per column.<br><br>b) Explain the handling of unstructured and semi-structured text data in Python. Write Python code to read a plain-text weather log file line by line using lazy evaluation (a generator), parse each line into a dictionary of fields (StationID, Temperature, Rainfall), and yield only valid records. | 3+3 | 1+2 |
| 3 | a) Explain how the requests library is used to call a REST API with authentication headers. Write Python code to retrieve hourly weather data from an API endpoint https://api.weather.example.com/hourly, passing an API key in the headers, and store the response in a Pandas DataFrame.<br><br>b) Explain lazy evaluation in Python and how generators support it. Write a Python generator function chunked_reader(filepath, chunksize) that reads a large weather CSV file in chunks using Pandas and yields one chunk at a time, computing the mean Temperature for each chunk. | 3+3 | 2 |
| 4 | a) Explain group-by aggregations in Pandas. How does groupby() differ from a pivot table when summarizing data?<br>b) Consider the following weather station dataset: | 2+4 | 3 |

| StationID | Region | Temperature | Humidity | Windspeed | Rainfall |
|---|---|---|---|---|---|
| W01 | North | 32 | 65 | 12 | NaN |
| W02 | South | NaN | 78 | 8 | 55 |
| W03 | North | 35 | NaN | 15 | 40 |
| W04 | East | 28 | 70 | NaN | 30 |
| W05 | South | NaN | 82 | 10 | 60 |
| W06 | East | 30 | NaN | 11 | 45 |

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| | Write Python code to: load the dataset, handle all missing values using an appropriate strategy for each column, and create a pivot table showing the average Rainfall per Region. | | |
| 5 | a) Explain three strategies for memory optimization in Pandas when working with large weather datasets. How do dtype downcasting and category types help?<br><br>b) Using the dataset from Question 4(b), write Python code to build a preprocessing pipeline that: encodes Region using label encoding, applies Min-Max scaling to Temperature, Humidity, and Windspeed, and outputs the transformed DataFrame. Implement the pipeline as a reusable function. | 2+4 | 3 |
| 6 | a) Explain regression analysis and trend analysis in the context of data science. What does the regression coefficient represent, and how is it interpreted?<br><br>b) Using the dataset from Question 4(b), write Python code using statsmodels.formula.api to fit a linear regression model with Temperature as the response variable and Humidity and Windspeed as predictors. Print the model summary and interpret the $R^2$ and coefficient values. | 2+3 | 4 |
| 7 | A meteorological department claims that the average daytime temperature in the region is 25°C. A random sample of 35 weather station readings produces the following statistics:<br>Sample mean = 27.4°C<br>Sample standard deviation = 4.8°C<br>Significance level = 0.05<br><br>Formulate the null and alternative hypotheses, compute the t-statistic manually, state the decision rule (t-critical for df = 34 at $\alpha = 0.05$ is $\pm 2.032$), and write Python code using scipy.stats.ttest_1samp to perform the test and print the p-value and conclusion. | 5 | 4 |
| 8 | a) Explain the role of Matplotlib subplots in comparative data analysis. What are the advantages of displaying multiple plots in a single figure?<br><br>b) Using the dataset from Question 4(b), write Python code using Matplotlib to create a figure with two subplots: a line plot showing Temperature across StationIDs on the left, and a scatter plot of Humidity versus Rainfall on the right. Handle missing values before plotting and add titles, axis labels, and a figure-level title. | 2+4 | 5 |
| 9 | a) Explain Seaborn's pair plot and its use in multivariate exploratory data analysis. Using the dataset from Question 4(b), write Python code to create a Seaborn pair plot for Temperature, Humidity, Windspeed, and Rainfall, with points coloured by Region.<br><br>b) Explain workflow automation using the schedule library in Python. Write Python code that uses schedule to run a weather data refresh function refresh_weather() every day at 6:00 AM, and a weekly summary function weekly_report() every Monday at 9:00 AM, keeping the scheduler running in a loop. | 3+3 | 5+6 |
| 10 | a) Explain data storage and retrieval strategies for data engineering pipelines. Compare CSV, Parquet, and SQLite in terms of read/write performance, compression, and suitability for pipeline stages. Write Python code to save a processed weather DataFrame as a Parquet file and reload it.<br>b) Write a CRON expression to trigger a Python script every hour and explain each field. What adjustments would you make to run it only during business hours (8 AM–6 PM, weekdays only)?<br><br>c) Using the dataset from Question 4(b), write Python code to generate an automated Excel report with two sheets: a Summary sheet containing mean, std, min and max for Temperature, Humidity, and Rainfall, and a Regional Averages sheet containing the average of all numeric columns per Region. | 3+2+3 | 6 |

| Exam | Regular | Full Marks | 60 |
|---|---|---|---|
| Level | BE | Pass Marks | 24 |
| Program | BCT | Time | 3 hrs. |
| Year/Part | III/I | | |

**TRIBHUVAN UNIVERSITY**
**INSTITUTE OF ENGINEERING**
**Examination Control Division**
Question Set: E

**Subject: Advanced Python Programming for Data Science (ENCT 325)**

- *Attempt All Questions*
- *Assume suitable data if necessary*

| QN | Questions | Marks | Units |
|---|---|---|---|
| 1 | **a)** Explain the difference between Python generators and iterators. Write a Python generator function sales_generator(records) that yields sales records one at a time from a list. Demonstrate how it can be used to compute cumulative revenue without loading all records into memory simultaneously.<br><br>**b)** Explain OOP in Python with a sales data science example. Write a SalesAnalyzer class that takes a Pandas DataFrame as input, with methods: total_revenue() that returns total revenue, top_region() that returns the region with the highest total revenue, and monthly_summary() that returns a grouped summary by Month. | 3+3 | 1 |
| 2 | **a)** Explain how lambda expressions simplify functional-style programming in Python. Using a list of sales dictionaries (each with keys Product, UnitsSold, Revenue), write Python code to: use map() to add a UnitPrice field (Revenue / UnitsSold) to each record, and use filter() to retain only records where UnitPrice > 2000.<br><br>**b)** Explain database access in Python using SQLite. Write Python code to: create a SQLite database sales.db, define a Sales table with columns SalesID, Region, Product, UnitsSold, Revenue, and Month, insert five sample records, and query all records where Region is 'North', ordered by Revenue descending. | 3+3 | 1+2 |
| 3 | **a)** Explain how APIs with pagination are handled in Python. Write Python code to retrieve all pages of sales data from a paginated API endpoint https://api.sales.example.com/records?page=1, incrementing the page parameter until no data is returned, and consolidate all records into a single Pandas DataFrame.<br><br>**b)** Explain chunked file reading in Pandas and when it is necessary. Write Python code to read a large sales CSV file in chunks of 500 rows and compute the total UnitsSold per Product across all chunks, accumulating results in a dictionary. | 3+3 | 2 |
| 4 | **a)** Explain multi-index DataFrames and the stack() and unstack() operations in Pandas. How does a multi-index pivot table differ from a single-index pivot table?<br>**b)** Consider the following sales dataset: | 2+4 | 3 |

| SalesID | Region | Product | UnitsSold | Revenue | Month |
|---|---|---|---|---|---|
| R01 | North | Laptop | NaN | 120000 | Jan |
| R02 | South | Phone | 45 | NaN | Jan |
| R03 | North | Laptop | 30 | 90000 | NaN |
| R04 | East | Tablet | NaN | 75000 | Feb |
| R05 | South | Phone | NaN | 60000 | Feb |

<table>
<tr><td></td><td><table><tr><td>R06</td><td>East</td><td>Tablet</td><td>22</td><td>NaN</td><td>Mar</td></tr></table>Write Python code to: load the dataset, handle all missing values using an appropriate strategy for each column type, and create a pivot table showing the average Revenue per Region and per Product.</td><td></td><td></td></tr>
<tr><td>5</td><td>a) Explain how categorical and time-series data are handled during preprocessing in Pandas. What issues arise with month-ordered categorical data and how are they resolved?<br><br>b) Using the dataset from Question 4(b), write Python code to build a preprocessing pipeline that: applies one-hot encoding to Region and Product, applies Min-Max scaling to UnitsSold and Revenue, and returns the transformed DataFrame. Implement as a callable pipeline function.</td><td>2+4</td><td>3</td></tr>
<tr><td>6</td><td>a) Explain EDA automation in Python. What is the benefit of automating descriptive analysis, and which libraries support it (e.g., pandas-profiling, ydata-profiling)?<br><br>b) Using the dataset from Question 4(b), write Python code to automate an EDA summary: compute descriptive statistics for UnitsSold and Revenue, compute and display the correlation matrix for all numeric columns, and print the count and percentage of missing values per column.</td><td>2+3</td><td>4</td></tr>
<tr><td>7</td><td>A regional sales manager claims that the average monthly revenue per sales record is NPR 80,000. A random sample of 32 sales records produces the following statistics:<br>Sample mean = 74,500<br>Sample standard deviation = 18,200<br>Significance level = 0.05<br><br>Formulate the null and alternative hypotheses, compute the t-statistic manually, state the decision rule (t-critical for df = 31 at $\alpha = 0.05$ is $\pm 2.040$), and write Python code using scipy.stats.ttest_1samp to perform the test and print the p-value and conclusion.</td><td>5</td><td>4</td></tr>
<tr><td>8</td><td>a) Explain the role of visualization in data storytelling. How do you choose between a bar chart, line chart, and histogram when analysing sales data?<br><br>b) Using the dataset from Question 4(b), write Python code using Matplotlib to create a figure with two subplots: a bar chart showing total Revenue per Region on the left, and a histogram showing the distribution of UnitsSold on the right. Handle missing values before plotting and add axis labels, titles, and a figure-level title.</td><td>2+4</td><td>5</td></tr>
<tr><td>9</td><td>a) Explain Plotly for interactive visualization. Using the dataset from Question 4(b), write Python code to create a Plotly interactive grouped bar chart showing total Revenue per Product for each Region, with hover information displaying UnitsSold.<br><br>b) Explain the design and implementation of an ETL pipeline in Python. Write Python code to implement a sales ETL pipeline that: extracts data from the dataset CSV, transforms by handling missing values, encoding Region and Product, and filtering records with Revenue above a threshold, and loads the clean output to a new CSV and a SQLite table.</td><td>3+3</td><td>5+6</td></tr>
<tr><td>10</td><td>a) Explain logging and monitoring in production data pipelines. Write Python code to implement a pipeline logger that: uses the logging module to log INFO messages at each ETL stage, logs WARNING if any column has more than 10% missing values, and logs ERROR if the output DataFrame is empty, saving all logs to pipeline.log.<br>b) Write a CRON expression to run a monthly sales report on the 1st day of every month at 9:00 AM. Explain each field and describe how you would schedule it on a Linux server.<br>c) Using the dataset from Question 4(b), write Python code to generate an automated Excel report with two sheets: a Summary sheet with descriptive statistics of UnitsSold and Revenue, and a Regional Performance sheet containing total Revenue and total UnitsSold grouped by Region and Month.</td><td>3+2+3</td><td>6</td></tr>
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